

URBANGAON AI LEGAL SEARCH PLATFORM

Complete User & Technical Documentation Manual — Enterprise Edition

Live MongoDB Atlas Cloud Integration Active

Version: 1.0.0 Enterprise Edition

Prepared For: AI Tool Development & Corporate Legal Team

Database: Live MongoDB Atlas Cloud Integration (ailegalsearchtool.zegxjbz.mongodb.net)

1. Overview & Architecture

The Urbangaon AI Legal Search Platform is an enterprise-grade AI legal intelligence system designed according to Module 2 of the Urbangaon AI Requirements Document. It consolidates Statutory Acts, Supreme Court & High Court Precedents, Corporate Contracts & MSAs, and YouTube Video Lectures into a single natural language search portal with exact traceability, video timestamp seeking, automated admin escalation, and a continuous knowledge improvement feedback loop.

System Architecture

```
graph TD
    User([Associate / Legal Researcher]) --> UI[Executive Web Portal UI]
    Contributor([Contributor / Paralegal]) --> UI
    Admin([Lead Counsel / Admin]) --> UI

    UI --> Server[Node.js / Express API Server]
    Server --> SearchEngine[Hybrid Vector & Semantic Legal Engine]
    Server --> IngestionPipeline[Multi-Modal Chunking & Ingestion Pipeline]
    Server --> AdminFeedbackDesk[Unanswered Resolution & Feedback Loop]
    Server --> ComplianceLogger[Immutable Audit & Analytics Engine]

    SearchEngine --> MongoAtlas[(MongoDB Atlas Cloud Database)]
    IngestionPipeline --> MongoAtlas
    AdminFeedbackDesk --> MongoAtlas
    ComplianceLogger --> MongoAtlas
```

2. Quick Start & Setup

Prerequisites

- Node.js (v18 or higher)
- NPM (v9 or higher)
- Internet connection for MongoDB Atlas Cloud sync

Environment Configuration (.env)

The application is configured in .env:

```
PORT=4000
MONGODB_URI=mongodb+srv://techdesk_db_user:aakash2899@ailegalsearchtool.zegxjbz.mongodb.net/urbangaon_legal_db?retryWrites=true&w=majority&appName=AILegalSearchTool
```

Starting the Server

Run the application using:

```
npm start
```

Open your browser and navigate to: <http://localhost:4000>

3. Detailed Feature-by-Feature User Guide

3.1 Header Controls & Role Switcher

Located at the top of the interface:

- Brand Logo & Live Indicator: Displays platform connectivity.
- Admin Desk Alert Badge: Displays the real-time count of pending low-confidence questions waiting for Lead Counsel review (e.g. Admin Desk: 3 Pending). Clicking it jumps directly to the Admin Desk.
- Active Role Switcher:
 - Employee / Associate: Read-only research access to Public and Internal documents; automatically routes low-confidence queries to admin.
 - Contributor / Paralegal: Can upload new legal documents, videos, and tag metadata.

- Legal Admin / Lead Counsel: Full administrative control resolves pending questions, feeds verified answers back into the knowledge base, manages documents, and inspects audit logs.

3.2 Tab 1: AI Legal Search & Research Workspace

This is the primary research workbench for lawyers, HR, and compliance teams.

1. Asking Natural Language Questions

- Type your question in plain English (or Hindi-English legal queries) into the central search bar.
- Example Queries:
 - "What is the notice period required to terminate an employee under the Industrial Disputes Act?"
 - "Why was Section 66A IT Act struck down by the Supreme Court?"
 - "What is the formula for 15 days severance pay in video lecture?"
 - "What is the quorum and composition for Internal Complaints Committee under POSH Act?"
 - "Are post-employment non-compete clauses valid under Indian Contract Act?"
 - "What is the service credit if uptime falls below 99.5% in Cloud SLA agreement?"
- Hit Enter or click Ask AI ->.

2. Quick Sample Query Pills

Click any sample pill below the search bar to run instant demonstration queries.

3. Faceted Filter Bar

Refine your search with dropdown filters:

- Document Type: All Types, Statutory Acts, Judgements, Contracts, YouTube Videos, or Admin Verified FAQs.
- Jurisdiction / Forum: All, India (Central), Supreme Court of India, Corporate Legal.
- Action Buttons:
 - [*] Save Query: Saves the query to your personal Research Vault (persisted in MongoDB Atlas).
 - [BRIEF] Export Legal Brief: Generates a formatted, printable legal research brief.

4. AI Legal Research Synthesis Card

The top answer card delivers:

- Synthesized Direct Legal Opinion: Synthesizes statutory sections, case precedents, or video transcripts into an executive summary.
- Confidence Indicator Badge:
 - [HIGH] High Confidence (>= 70%): Verified authority match.
 - [MEDIUM] Medium Confidence (45% - 69%): Cautionary preliminary response.
 - [LOW/ESCALATED] Low Confidence (< 45%) / Gaps: Transparent notice "No confident match found" + automatically escalates query to the Admin Desk.
- Statutory Legal Disclaimer: Mandatory legal research aid disclaimer.
- Toolbar Actions:
 - [COPY] Copy Synthesis: Copies the answer text to clipboard.
 - [*] Bookmark Clause: Saves the top statutory clause to your vault.

5. Exact Source Citations & Authorities List

Each retrieved source is presented as a card with:

- Document Type tag ([ACT] Act, [CASE] Judgement, [CONTRACT] Contract, [VIDEO] YouTube Video).
- Court / Forum and Year.
- Section / Paragraph badge (e.g. Section 25F - Conditions precedent to retrenchment).
- Supersession Warning Banner: Displays red alert if the law was amended or struck down (e.g. Section 66A IT Act in Shreya Singhal v. UOI).
- Exact quoted statutory excerpt.
- Action Buttons:
 - [SEARCH] Inspect Full Text: Opens the Full Document Inspector Modal with highlighted chunks.
 - [PLAY] Play at Timestamp [MM:SS] (For YouTube Videos): Opens the video player and seeks directly to the exact second of that legal point.

6. Right Sidebar: Related Queries & Personal Vault

- Related Legal Queries: Contextual suggestions based on legal doctrine ("Users who searched this also viewed..."). Click any suggestion to search instantly.
- [*] Saved Clauses & Queries: Displays all your bookmarked clauses and saved queries. Click any saved item to re-execute.

3.3 Tab 2: Knowledge Base & Ingestion Repository

Central library of all indexed legal authorities.

1. Browsing & Category Filtering

Filter documents using category pills:

- All Categories, Labour & Employment, Cyber & Tech Law, Constitutional & Privacy Law, Corporate Law, Commercial & Contract Law, Video Webinars.

2. Inspecting Indexed Chunks

Click View Chunks on any document card to open a modal listing all section chunks, paragraph breakdowns, and tags.

3. Ingesting New Legal Material (Ingest New Content / Video)

Click the gold button to open the Ingestion Modal:

1. Drag & Drop / File Browser:
 - Click the blue dashed box or drop .pdf, .docx, .txt, .md, .json, .csv, or scanned files.
 - The system extracts text, auto-populates the Title, and prepares it for chunking.
2. Metadata Fields:
 - Document Title: e.g., Digital Personal Data Protection Act, 2023.
 - Document Type: Act, Judgement, Contract, YouTube Video, or Internal Advisory.
 - Jurisdiction & Court: e.g., India (Central), Supreme Court of India.
 - Category: Dropdown category.
 - Confidentiality Tier:
 - Public: Searchable by all employees.
 - Internal: Company employees only.
 - Confidential: Restricted to Lead Counsel & Admin.
 - Year: e.g., 2024.
3. YouTube Video Ingestion Fields:
 - Set Document Type to "YouTube Video".
 - Enter YouTube Video URL (e.g. <https://www.youtube.com/watch?v=dQw4w9WgXcQ>).
 - Enter Duration (e.g. 18:45).
 - In the text box, paste the transcript with [MM:SS] timestamps:


```
```text
[02:10] Code for Independent Directors (Schedule IV): Explaining mandatory meetings...
[07:45] Board Committee Requirements: Audit committee composition...
[12:30] Liability & Safe Harbour Protections: Section 149(12) rules...
```
```
4. Superseded Law Toggle:
 - Check "Mark as Superseded / Struck Down Law" if the document has been repealed, amended, or overturned by a higher court.
5. Click Chunk & Ingest Document ->:
 - The system automatically parses sections and timestamps into distinct chunks, saves them to MongoDB Atlas, and makes them searchable instantly!

3.4 Tab 3: Admin Resolution Desk & Dynamic Feedback Loop

Solves the cold-start and legal knowledge gap problem.

1. Unanswered & Low-Confidence Queue

- When any user asks a question that receives low confidence (< 45%), the engine logs the question and displays it here.
- Sorted by frequency (e.g. [PRIORITY] 7x Asked, [PRIORITY] 4x Asked), prioritizing the most urgent organizational legal questions.

2. Resolving a Question & Retraining the Knowledge Base

1. Click [ADMIN] Resolve & Feed KB on any pending question.
2. Enter the official verified legal guidance or statutory interpretation.
3. Keep "Feed directly into AI Knowledge Base" checked.
4. Click Publish Resolution & Update KB ->.
5. Immediate Result:
 - The question status changes to Resolved.
 - A new verified chunk is created in MongoDB Atlas.
 - When any user searches that query again, the engine instantly returns the verified answer with High Confidence!

3.5 Tab 4: Executive Analytics & Gap Insights

Provides real-time visibility into organization-wide legal research activities:

- KPI Metrics:
 - Total Legal Searches Logged
 - Average Confidence Score (%)
 - Pending Legal Gap Questions
 - Verified Admin Feedback Loops Integrated
- Knowledge Base Category Distribution Chart: Visual breakdown of document coverage.
- Search Confidence & Escalation Distribution: Breakdown of High, Medium, and Escalated queries.

3.6 Tab 5: Compliance & Immutable Audit Trail

Regulatory compliance log tracking every search and action:

- Timestamp: Exact UTC and local execution time.
- User & Role: Employee name and active access tier.
- Action Type: SEARCH_QUERY, UNANSWERED_ESCALATION, DOCUMENT_INGESTION, ADMIN_FEEDBACK_RESOLVED.
- Query / Activity: Full text of the query.

- Confidence Tier: High, Medium, Low (Routed to Admin).
- Retrieved Documents: List of all cited authorities.
- IP Address: Client network address.

4. REST API Reference

| Endpoint | Method | Description | Sample Request / Parameters |
|-----------------------------------|------------|---------------------------------------|--|
| /api/search | POST | Execute natural language legal search | { "query": "Notice period under IDA?", "filters": {}, "userRole": "Employee" } |
| /api/documents | GET | Retrieve accessible documents | Query params: userRole=Employee, category=Corporate |
| /api/documents/:id | GET | Get full document details and chunks | Param: id |
| /api/documents | POST | Ingest new document / video | Multipart or JSON payload with title, docType, rawContent |
| /api/documents/:id | PUT | Update document metadata | Param: id, body: { "isSuperseded": true } |
| /api/documents/:id | DELETE | Delete document (Admin only) | Param: id |
| /api/admin/unanswered | GET | List unanswered questions queue | None |
| /api/admin/unanswered/:id/resolve | POST | Resolve question and feed KB | { "adminAnswer": "Verified text...", "addToKnowledgeBase": true } |
| /api/admin/unanswered/:id/dismiss | POST | Dismiss question | { "note": "Out of scope" } |
| /api/analytics | GET | Get executive analytics & charts | None |
| /api/audit-logs | GET | Get compliance audit logs | None |
| /api/bookmarks | GET / POST | Manage user bookmarks vault | { "title": "Sec 25F", "docTitle": "IDA 1947", "excerpt": "..." } |
| /api/saved-searches | GET / POST | Manage saved search queries | { "query": "POSH ICC committee rules" } |

5. MongoDB Atlas Database Schema Reference

The platform connects to MongoDB Atlas under database urbangaon_legal_db:

1. legaldocuments Collection

```
{
  "_id": "ObjectId(...)",
  "id": "doc-act-001",
  "title": "Industrial Disputes Act, 1947",
  "docType": "Act",
  "jurisdiction": "India (Central)",
  "court": "Parliament of India",
  "year": 1947,
  "status": "Active",
  "confidentiality": "Public",
  "category": "Labour & Employment Law",
  "isSuperseded": false,
  "chunks": [
    {
      "id": "chunk-act-001-1",
      "section": "Section 25F",
      "heading": "Conditions precedent to retrenchment of workmen",
      "content": "No workman employed in any industry...",
      "timestampSeconds": null,
      "tags": ["retrenchment", "notice period", "workman"]
    }
  ]
}
```

2. unansweredquestions Collection

```
{
  "_id": "ObjectId(...)",
  "id": "unans-001",
  "query": "What is the penalty for non-compliance with DPDP Act 2023?",
  "askedBy": "Rohan Verma (Employee)",
  "frequency": 7,
  "status": "Pending Review",
  "category": "Data Privacy",
  "confidenceScore": 0.18
}
```

3. auditlogs Collection

```
{
  "_id": "ObjectId(...)",
  "id": "audit-1772600000",
  "timestamp": "2026-09-02T10:00:00.000Z",
  "userName": "Adv. Rajesh Sharma (Lead Counsel)",
  "userRole": "Admin",
  "action": "SEARCH_QUERY",
  "query": "What is the notice period for workman retrenchment?",
  "confidenceScore": 0.94,
  "confidenceTier": "High",
  "retrievedDocs": ["Industrial Disputes Act, 1947"],
  "ipAddress": "127.0.0.1"
}
```

6. Verification & Automated Test Suites

The repository contains two automated test suites:

1. Comprehensive System Test Suite

Runs 19 end-to-end test cases covering statutory search, video timestamps, supersession alerts, admin escalation, resolution feedback loop, document ingestion, and audit logging:

```
node tests/testSuite.js
```

2. Live Ingestion & Multi-Modal Search Test

Performs real-time ingestion of Acts, Judgements, Contracts, and YouTube Videos with timestamp seeking directly into MongoDB Atlas:

```
node tests/liveIngestionTest.js
```

Urbangaon AI Legal Search Platform Empowering Corporate Legal & Compliance with Precision Intelligence.